Emily Appleby

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EDUCATION

University of Colorado Boulder — Boulder, CO
Bachelor of Science in Mechanical Engineering & Minor in Economics — May 2027
Relevant Coursework: *Component Design, System Dynamics, Advanced Computer Vision, Sustainable Energy, Introduction to Econometrics*

PROFESSIONAL EXPERIENCE

Cornell Tech — New York City, NY
Summer Intern, Interaction Research Lab — May 2026 - August 2026

- Executed 16 campus café deployments of the trash barrel robot using Wizard-of-Oz control and Vision Language Models (VLMs) for social intelligence, logging 500+ human-robot interactions
- Analyzed deployment data to evaluate VLM-driven social behaviors, noting errors and user reactions
- Troubleshot robot bases, resolved wiring issues, and swapped the power supply to run on standard drill batteries for reliability
- Learned and applied Robot Operating System (ROS) 1 and 2
- Managed hardware inventory to support Fall coursework

Reon Technology — Chelmsford, MA
Mechanical Engineering Intern — May 2025 - August 2025

- Built a lighter demonstration version of the mobile power station using a smaller battery and custom 3D-printed parts
- Handled quick mechanical fixes, including 3D-printing a custom hinge for shipping, along with field software and hardware updates
- Authored standard operating procedures (SOPs) for assembling MC4 connectors and servicing the mobile power station

Valley Pond — Lincoln, MA
Lifeguard — August 2022 - September 2024

- Monitored swimmer safety and managed emergency response protocols

ENGINEERING EXPERIENCE

Drill-Powered Vehicle Design Project — Boulder, CO
Team Member — August 2025 - December 2025

- Contributed to CAD development and performed finite element analysis (FEA) on the vehicle frame
- Assisted with manufacturing, vehicle assembly, and getting the vehicle to run the obstacle course on test day

Electric conversion of 1986 Porsche 944 — Lincoln, MA
Team Member — April 2023 - Present

- Dismantled a wrecked Tesla Model 3 to salvage components, repurposing the battery modules for a home storage battery
- Collaborated on the ongoing mechanical and electrical integration for the conversion

LEADERSHIP & SERVICE

CU Boulder Nordic Ski Club, Officer — November 2023 - Present

TECHNICAL SKILLS

CAD: SolidWorks (Certified Design Associate), OnShape (100hrs), PTC Creo (1.5 years)

Programming Languages: Python, C, Java, MATLAB, HTML, CSS, JavaScript, SQL

Fabrication: Lathes, Mills, 3D-printers, and Laser Cutters